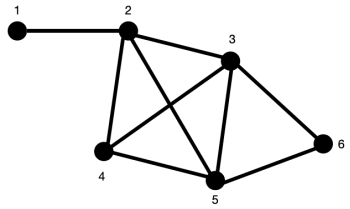
CSCI-570 Homework 7

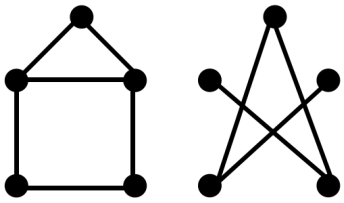
Due Date: Tuesday, December 2nd, 11:59pm

Problem 1.
A *clique* in a graph $G = (V, E)$ is a subset of the vertices $C \subseteq V$, where every two vertices are connected by and edge, i.e. for every $u, v \in C$, $(u, v) \in E$. The following decision question can be asked: Given a graph G and an integer k, does G have a clique of size at least k?



For example, in the graph above vertices $\{2, 4, 5\}$ and $\{3, 5, 6\}$ form cliques of size 3 (and there are more of the same size), while the largest clique is of size 4 consisting of $\{2, 3, 4, 5\}$.

The *complement* of a graph $G = (V, E)$ is another graph $\overline{G} = (V, E')$ with the same set of vertices, but in $\overline{G}$ exactly those $u, v \in V$ vertices are connected by a $(u, v) \in E'$ edge, which are not connected in G, that is, $(u, v) \notin E$. (Thus $E' = \{V \times V \setminus E\}$.)



G and its complement $\overline{G}$

Prove the following statements:

(a) Let $G = (V, E)$ be a graph and $\overline{G}$ its complement. A subset $S \subseteq V$ of the vertices is an independent set in G if and only if it forms a clique in $\overline{G}$.

(b) The decision problem *clique* as formulated above is an $\mathcal{NP}$*-complete* problem.

[4+8 points]

Problem 2.
The $Half\text{-}Clique$ decision problem is as follows: given a graph G with $|V| = n$ vertices, does it contain a *clique* with at least $\frac{n}{2}$ nodes? Show that $Half\text{-}Clique$ is $\mathcal{NP}$*-complete* using the $\mathcal{NP}$*-completeness* of *Clique*.

[8 points]

Problem 3.
Consider an instance of the *Satisfiability Problem*, specified by clauses $C_1, ..., C_k$ over a set of Boolean variables $x_1, ..., x_n$. We say that the instance is *monotone* if each clause contains only non-negated variables; that is, each term is equal to x_i, for some i, rather than $\overline{x_i}$. Monotone instances of *Satisfiability* are very easy to solve: they are always satisfiable by setting each variable equal to 1.

For example, suppose we have $(x_1 \vee x_2)$ & $(x_1 \vee x_3)$ & $(x_2 \vee x_3)$. This is monotone, and indeed the assignment that sets all three variables to 1 ("True") satisfies all the clauses. But we can observe that this is not the only satisfying assignment; we could also have set x_1 and x_2 to 1, and x_3 to 0. Indeed, for any monotone instance, it is natural to ask how few variables we need to set to 1 in order to satisfy it.

Given a monotone instance of *Satisfiability*, together with a number k, the decision problem *Monotone Satisfiability with Few True Variables* asks: Is there a satisfying assignment for the instance in which at most k variables are set to 1? Prove this problem is $\mathcal{NP}$*-complete*.

[10 points]